- **Stability:** This configuration minimizes the impact of internal parasitic capacitance and ensures a stable resistance reference during the high-speed binary search process.

3.5.4 Resolution and Error Analysis

The resolution of the ohmmeter is defined by the 128 discrete steps of the 10 kΩ digital potentiometer:

$$Resolution = \frac{10000\Omega}{128\ steps} = 78.125\Omega/step$$

Because the "balance point" may fall between two discrete steps, the system carries an inherent quantization error. To maintain engineering transparency, the software appends a plus or minus 36 Ω warning to the measurement displayed on the LCD, ensuring the user is aware of the hardware's precision limits.

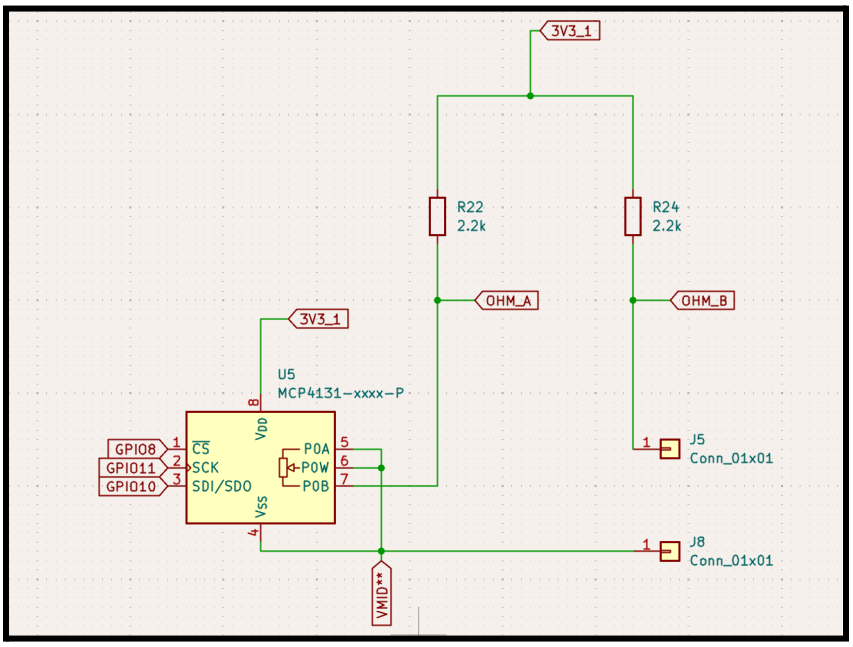


Figure 3.8 - Ohmmeter Schematic